

Mucilage Forecast for the North Adriatic Sea Coast

A Functional Data Analysis approach for short-term algal bloom prediction

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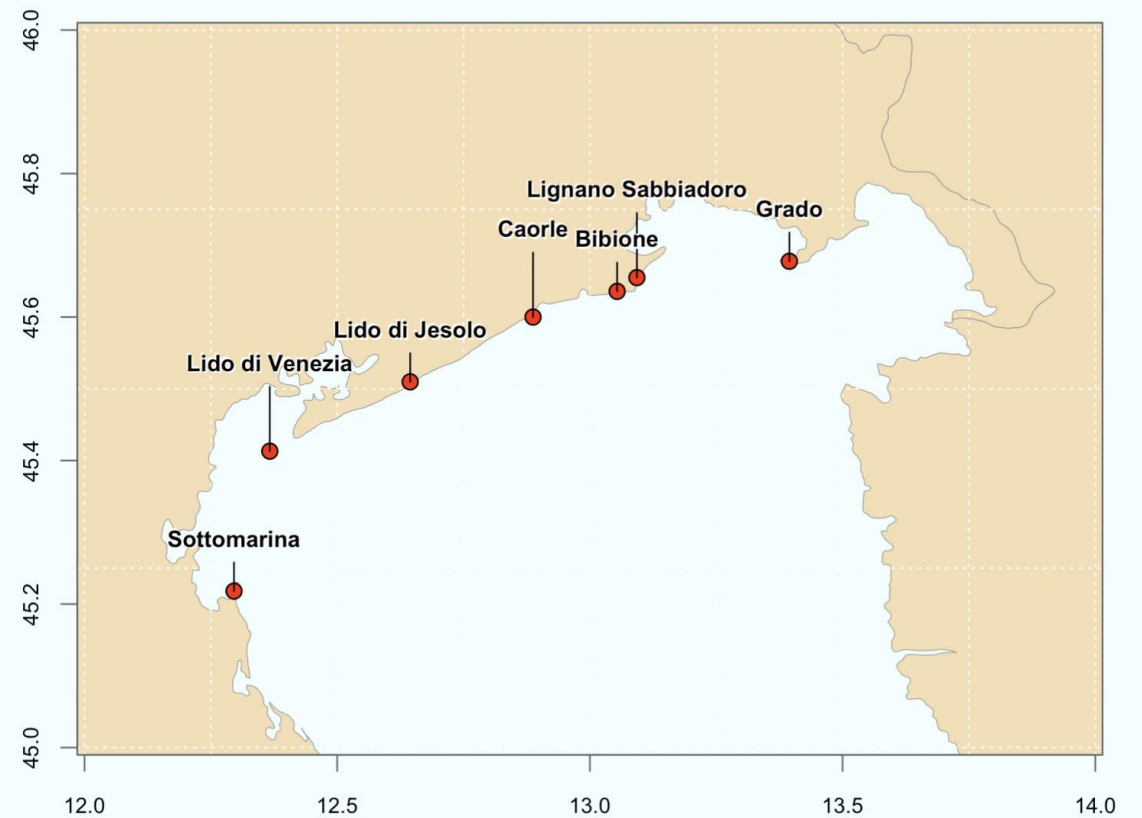
PROJECT GOAL

Northern Adriatic Sea periodically suffers from mucilage phenomena which negatively impacts the tourist experience.

Meteo.Adriatico commissioned us a project to develop a data-driven predictive tool.

Our goal is to provide beach enjoyers a short-term forecast to identify beaches with the clearest water.

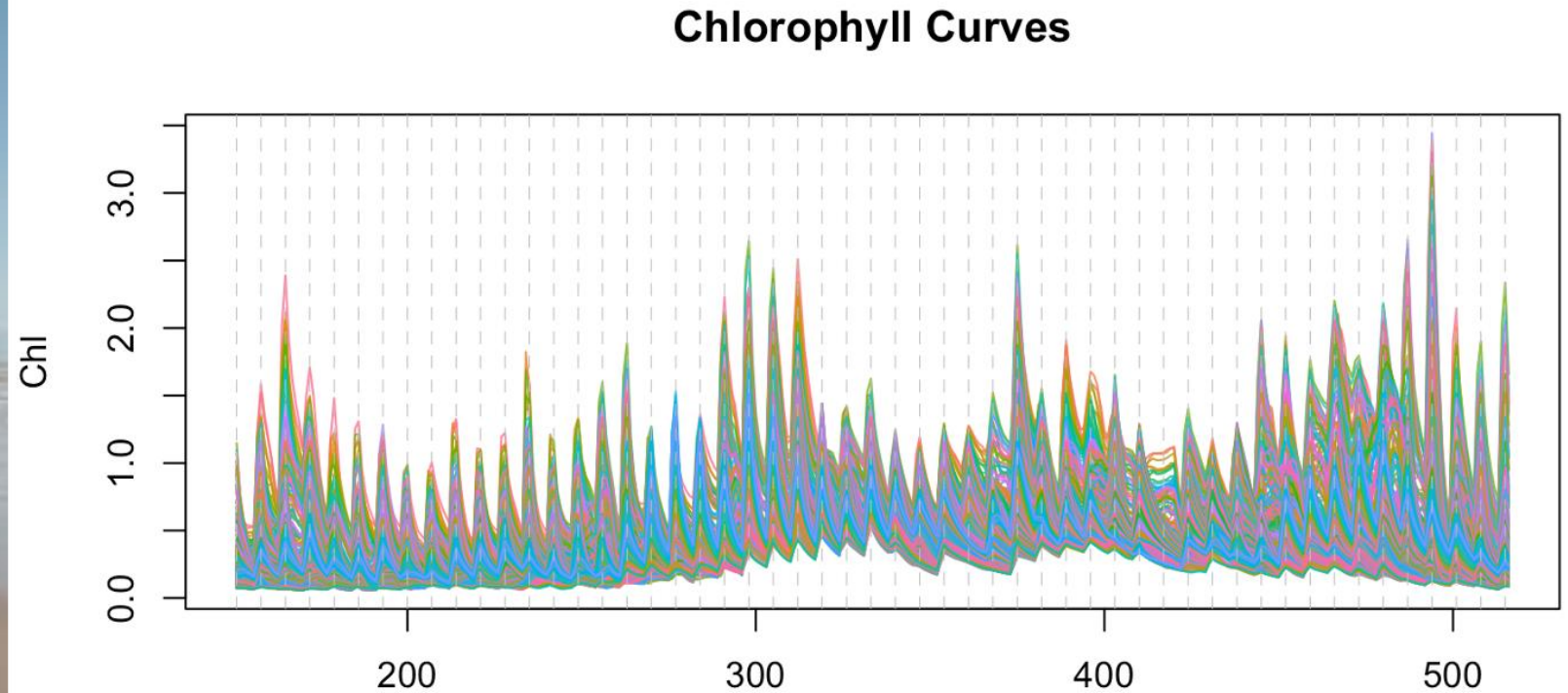
Seaside Resorts of North Adriatic Sea



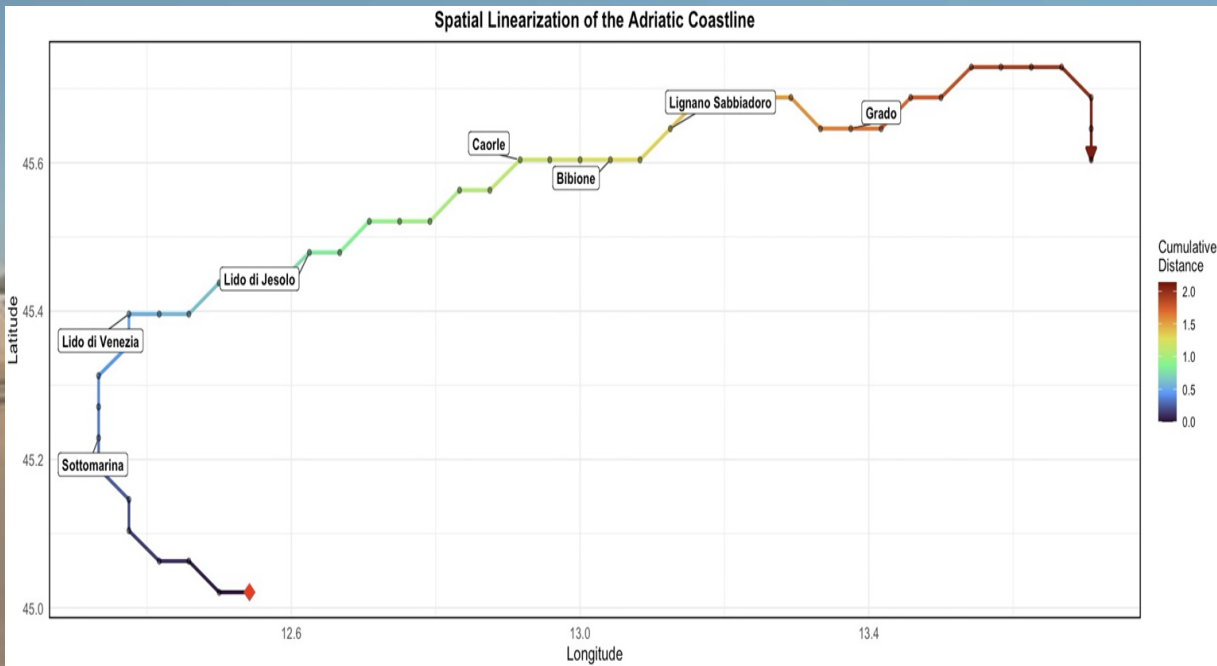
DATA SOURCE & RELIABILITY

Target Variable: Chlorophyll-a, daily measurements over the span of three years.

The "Weekly Artifact":
Numerical 7-day oscillation found in satellite data assimilation.



COAST LINEARIZATION: FROM 2D TO 1D



Complex 2D coordinates are computationally inefficient for precise spatial modeling.

- **Concave Hull:** Exact boundary extraction of the coastline.
- **Nearest Neighbor:** Logic ordering from Po Delta to Grado
- **Cumulative Distance:** Transformed coast into a 1D scalar metric

SMOOTHING VIA B-SPLINES

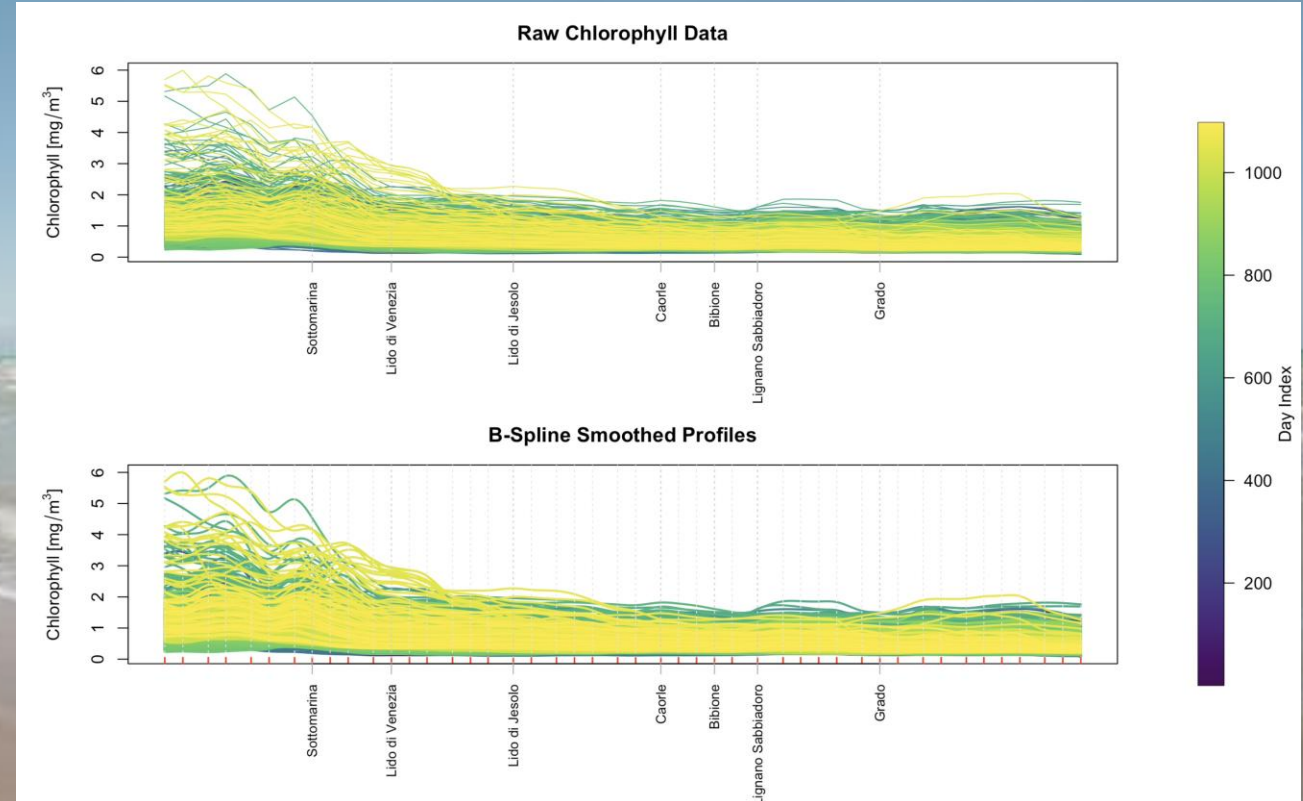
Raw satellite data is inherently discrete and spatially noisy.

We transform point data into continuous data.

- Cubic B-Splines for continuous spatial functions.
- Knots at every measured position along the coast.
- Roughness penalty for curvature controlling.

Minimize Penalized Sum of Squares Error:

$$PENSSE_{\lambda} = \sum_{i=1}^N [y_{t,i} - X_t(d_i)]^2 + \lambda \int [X_t''(d)]^2 dd$$



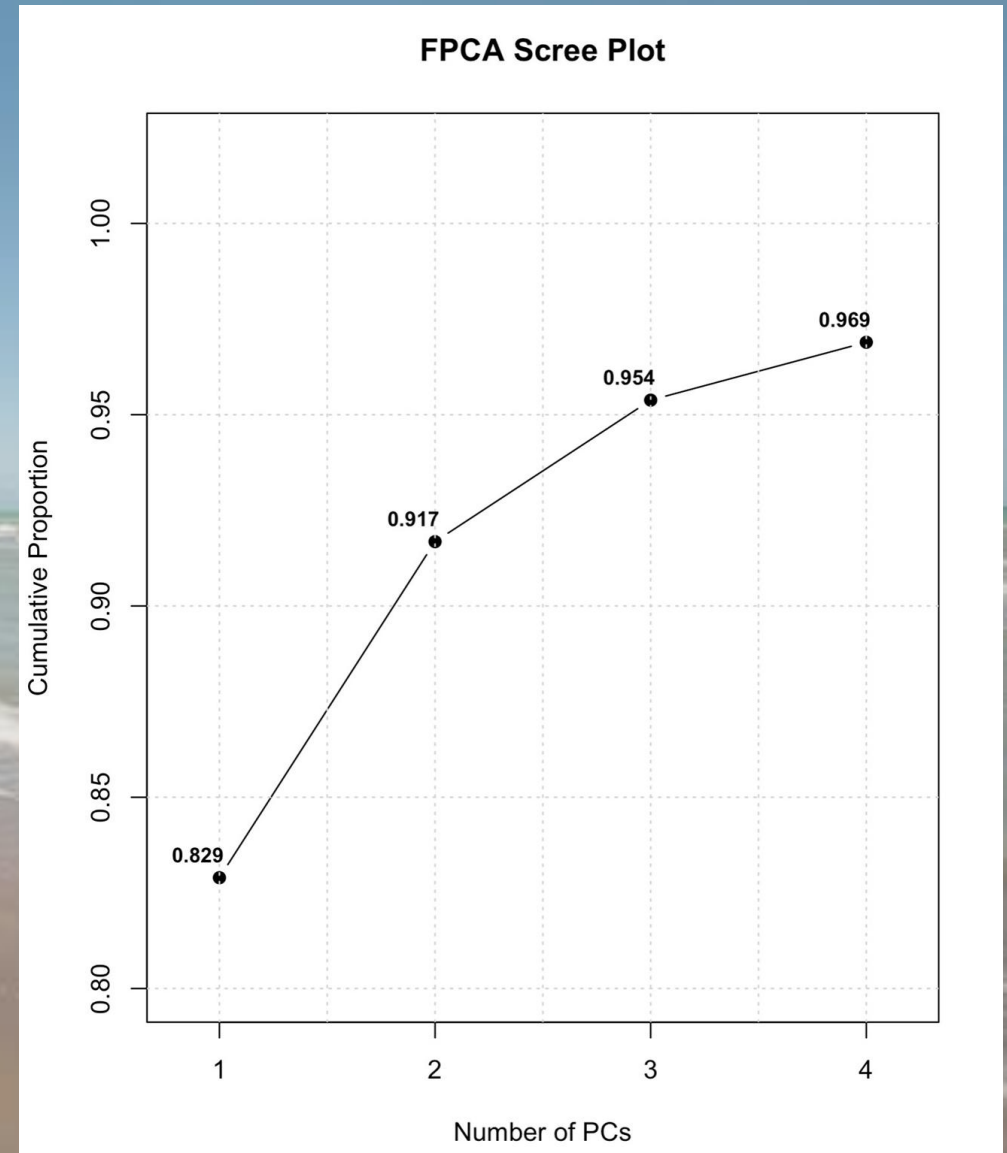
COMPRESSING THE COASTLINE (FPCA)

Functional PCA

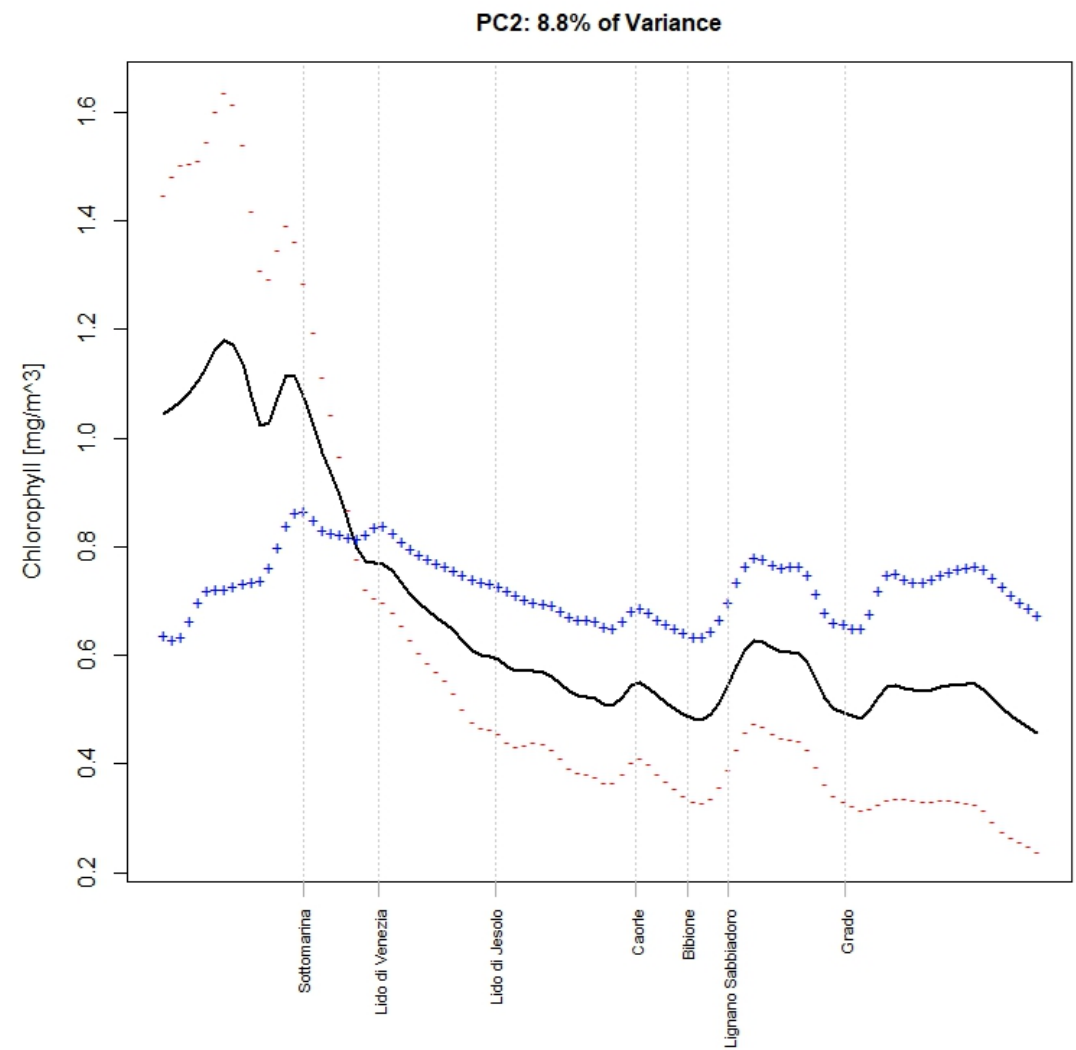
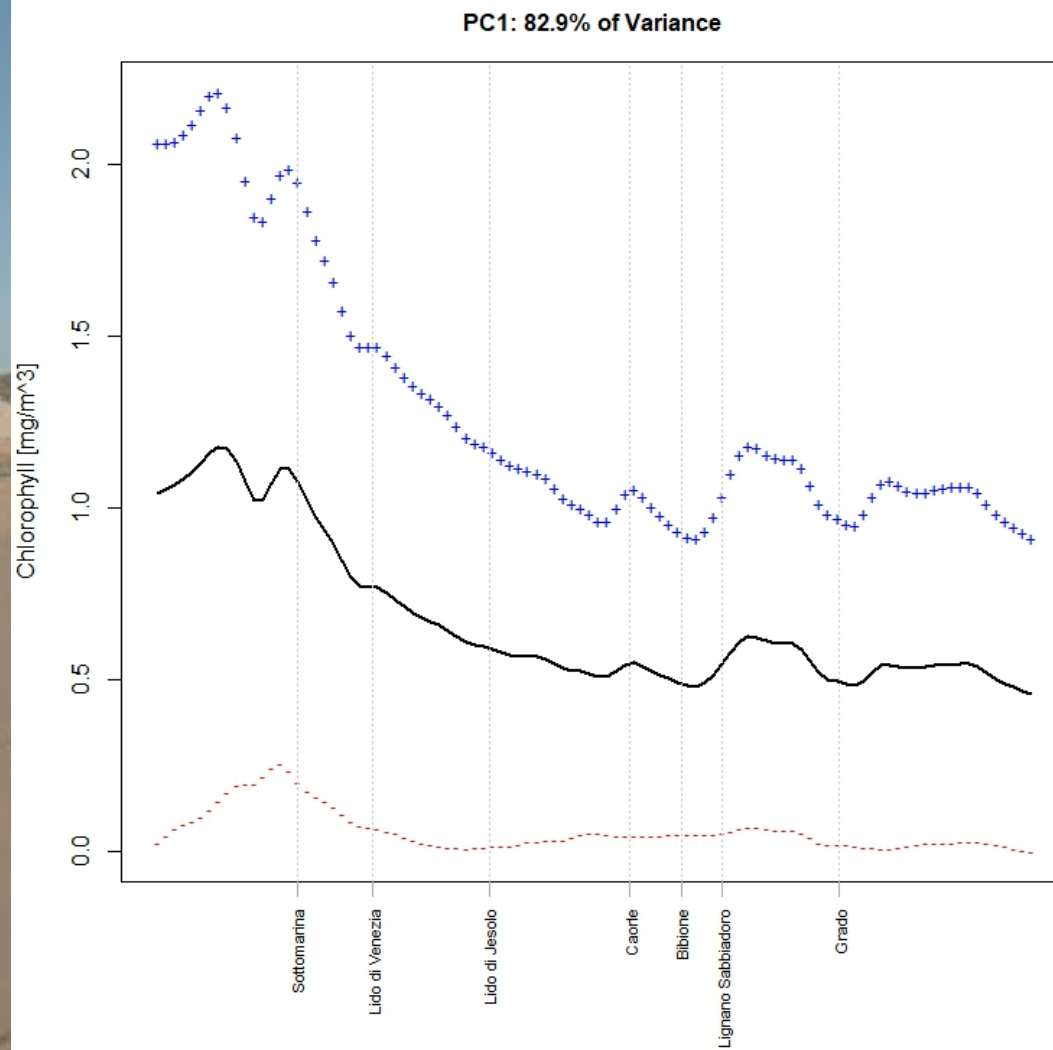
FPCA computed on the year previous the prediction date, reduces the scores of the Spline Basis into 4 components.

$$X_t(d) \approx \mu(d) + \sum_{j=1}^4 \xi_{t,j} \psi_j(d)$$

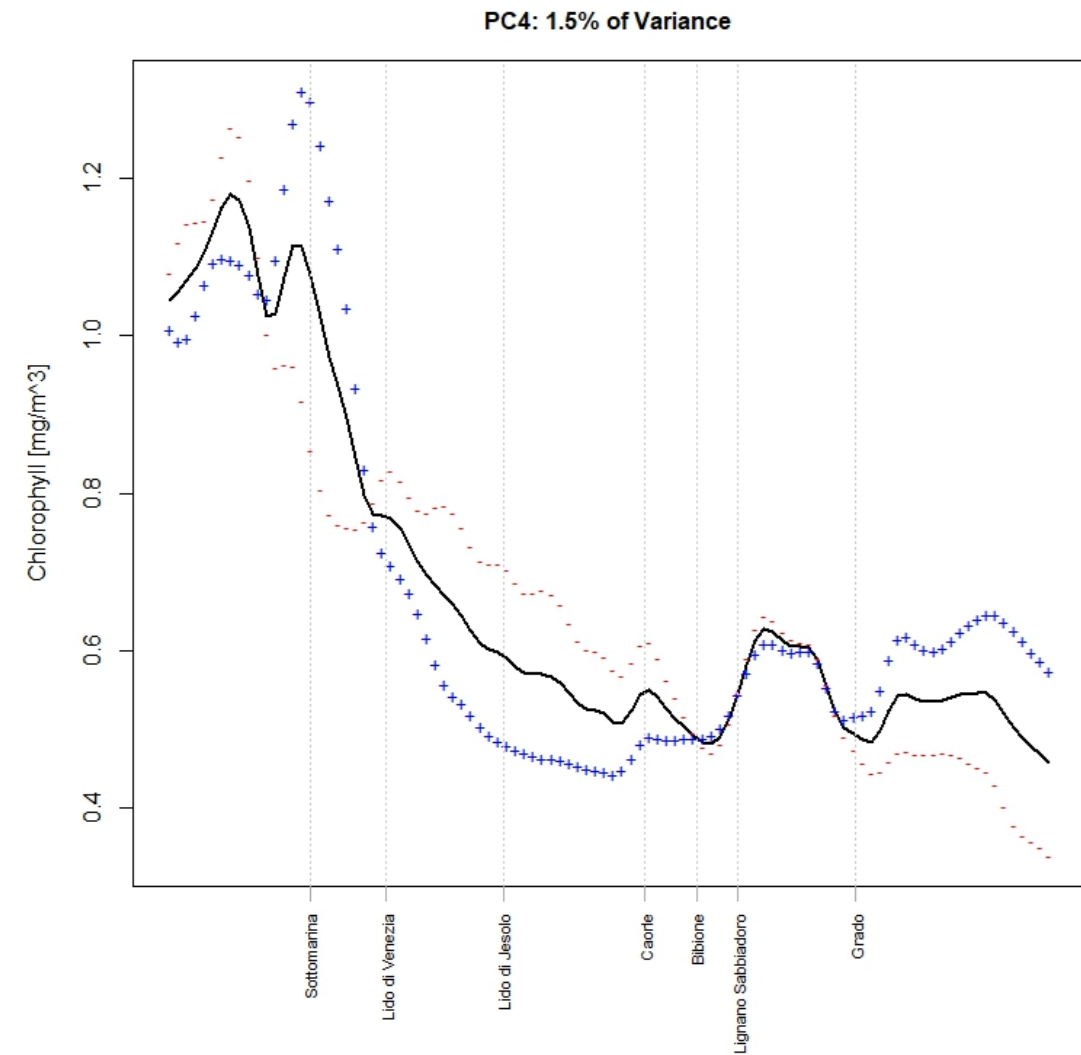
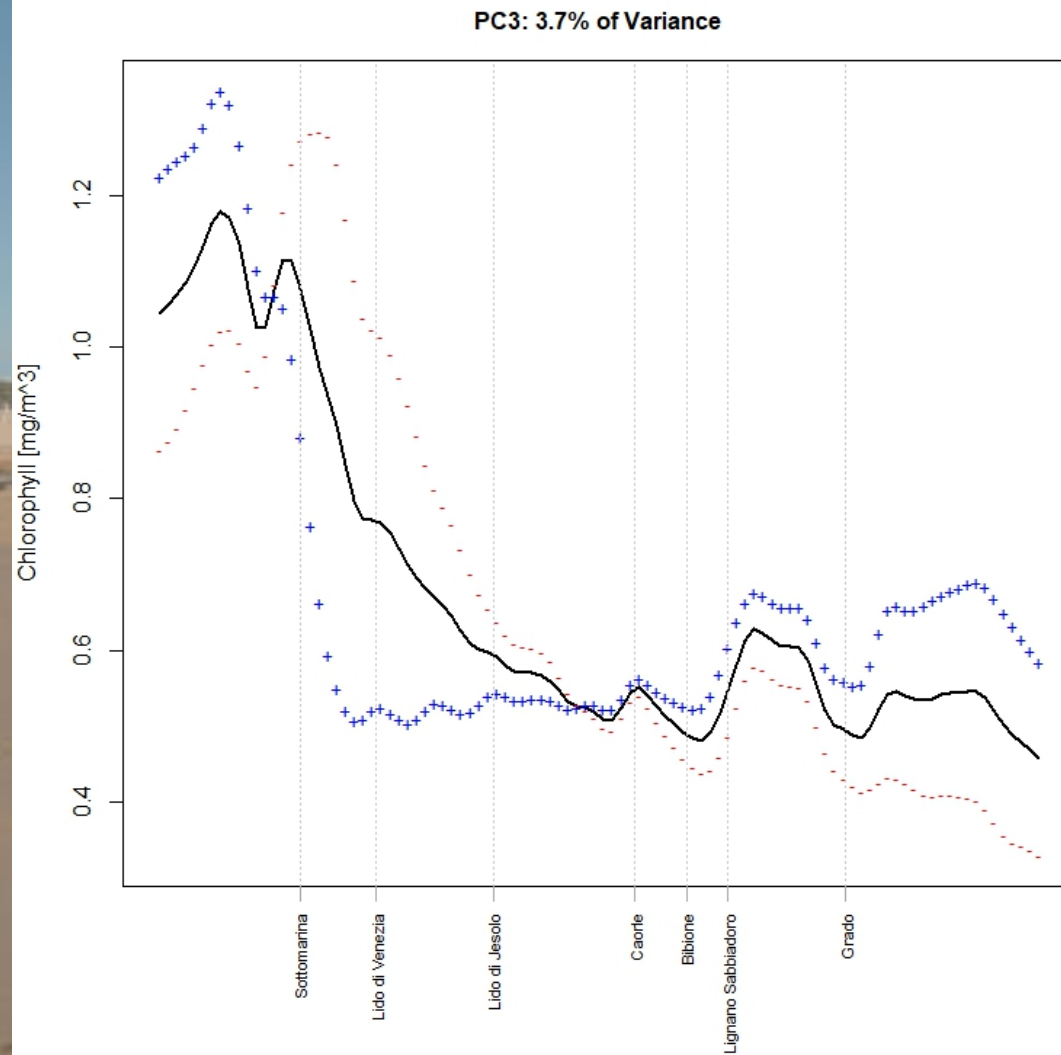
The first 4 Components explain >96% of total spatial variance.



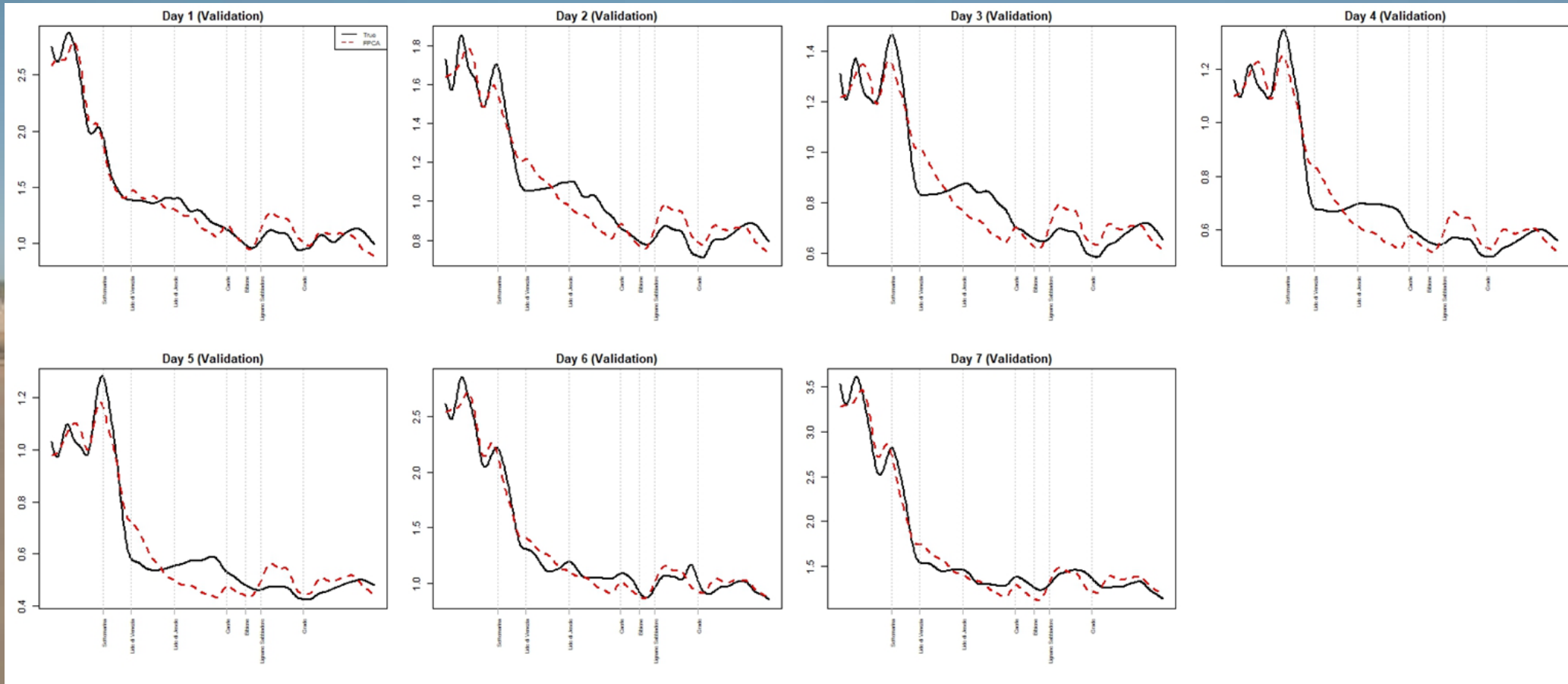
VISUALIZING PRINCIPAL COMPONENTS



VISUALIZING PRINCIPAL COMPONENTS



Validation projected on the 4 PC



MULTIVARIATE TIME-SERIES MODELS

XGBOOST

ML approach. It operates as an **ensemble of Classification and Regression Trees (CART)**, where new trees are added sequentially to correct the errors of existing trees.

VAR

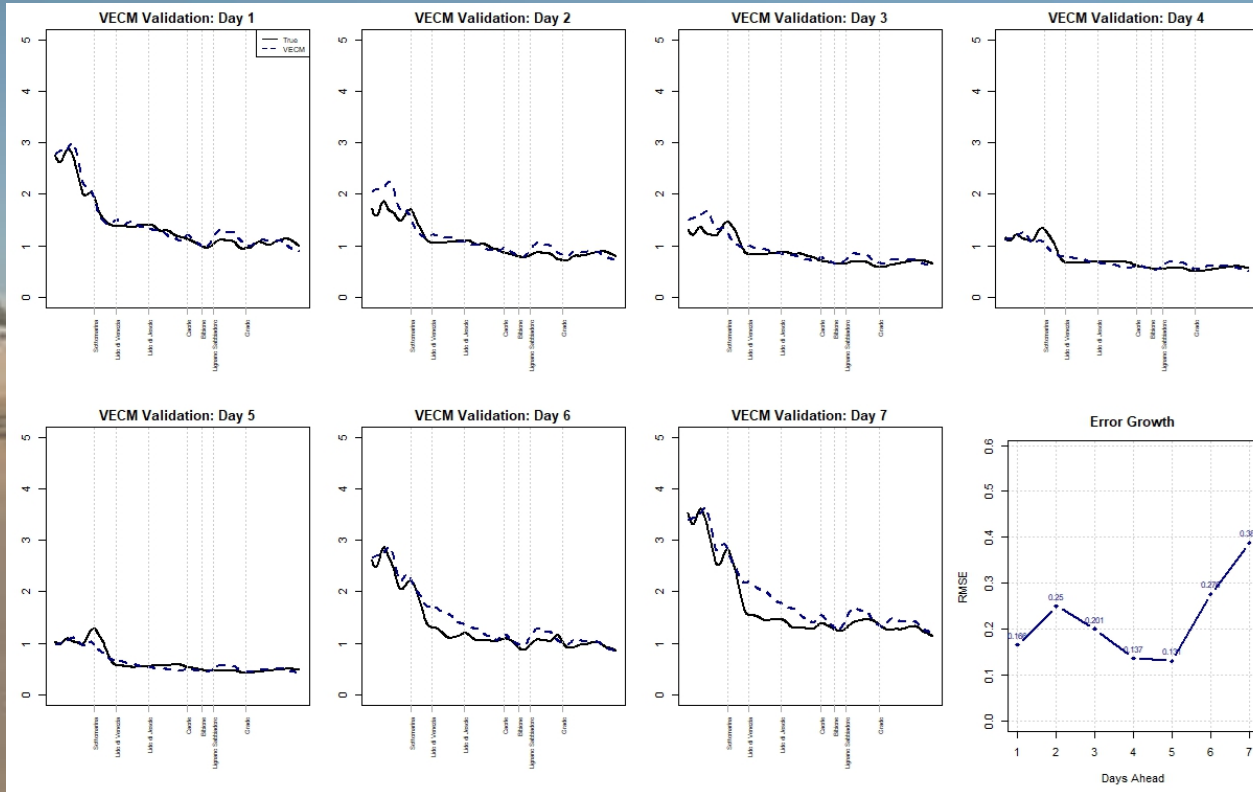
The Vector Autoregression (VAR) is a stochastic process that generalizes the univariate Autoregressive (AR) model to a multivariate framework.

VECM

WINNER! Restricted form of VAR designed for use with **non-stationary** series that are known to be **cointegrated**.

$$\beta' \xi_t = \epsilon_t$$

MODEL PERFORMANCE (7-DAY HORIZON)



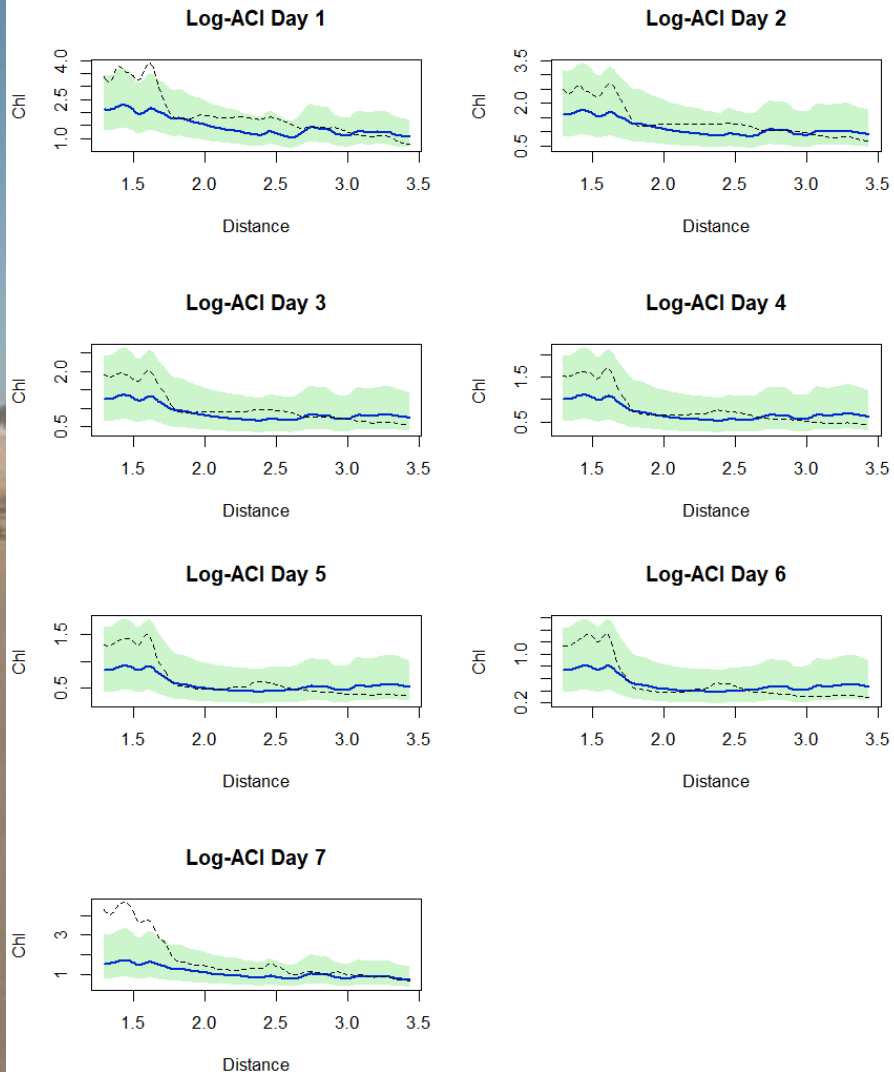
VECM: Vector Error Correction Model

restricted form of a Vector Autoregression (VAR)
designed for use with non-stationary series.

$$\Delta \xi_t = \alpha(\beta' \xi_{t-1}) + \sum \Gamma_i \Delta \xi_{t-i} + \epsilon_t$$

Key Feature: Distinguishes between short-term shocks and long-term equilibrium adjustments.

PREDICTION



Beyond point estimates

GOAL: Make risk assessment with a given error

FEATURES:

- ❖ Physical validity
- ❖ Accounts for the time-dependency

DISCUSSION

Results

- ❖ Interdependence of Spatial Zones
- ❖ Predictive stability
- ❖ Reliability of the Model

Future developments

- ❖ Point estimate with Ensemble methods or NNs
- ❖ Physics-Informed Model